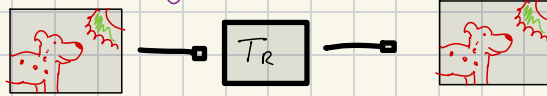


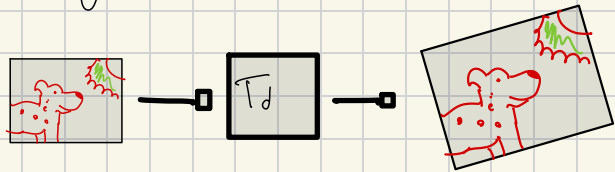
Image Manipulation

① Image Filtering: mudança no **Range**

$$g(x,y) = T_R(f(x,y))$$



② Image Warping: mudança no **domain**

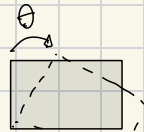


$$g(x,y) = f(T_d(x,y))$$

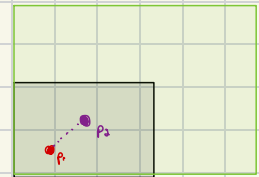
★ Global Transformations



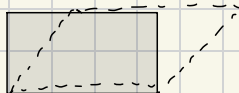
Translation



Rotation

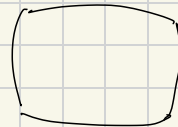


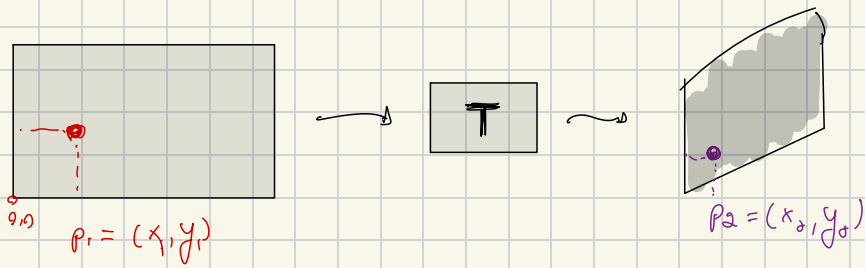
Scaling



Affine

...

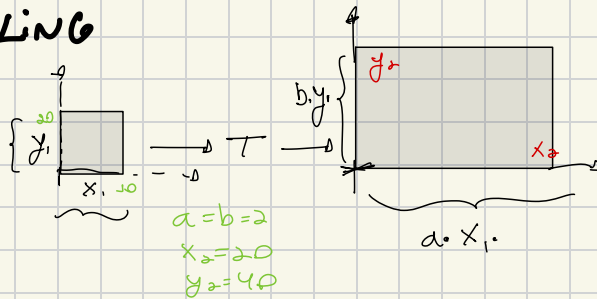




$$p_2 = T \cdot p_1$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

0 SCALING



$$\begin{aligned} a &= b = 2 \\ x_2 &= 20 \\ y_2 &= 40 \end{aligned}$$

$$\begin{aligned} x_2 &= a \cdot x_1 \\ y_2 &= b \cdot y_1 \end{aligned}$$

① Forward

$$\begin{aligned} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} &= S \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \begin{bmatrix} a & 0 \\ 0 & b \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = \\ &= \begin{bmatrix} a \cdot x_1 + 0 \cdot y_1 \\ 0 \cdot x_1 + b \cdot y_1 \end{bmatrix} = \\ \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} &= \begin{bmatrix} a x_1 \\ b y_1 \end{bmatrix} \end{aligned}$$

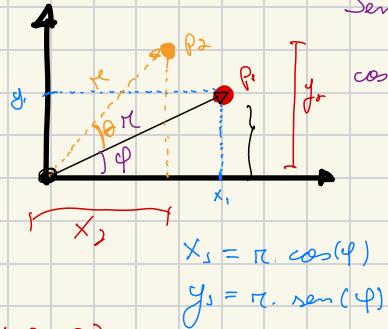
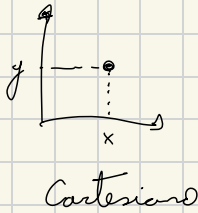
② Inverse

$$x_1 = \frac{1}{a} x_2$$

$$y_1 = \frac{1}{b} y_2$$

$$\begin{bmatrix} x_1 \\ y_1 \end{bmatrix} = S^{-1} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} \frac{1}{a} & 0 \\ 0 & \frac{1}{b} \end{bmatrix} \begin{bmatrix} x_2 \\ y_2 \end{bmatrix}$$

★ Rotation



$$x_2 = r \cdot \cos(\phi + \theta)$$

$$y_2 = r \cdot \sin(\phi + \theta)$$

$$\sin(\alpha + \beta) = \sin \alpha \cdot \cos \beta + \sin \beta \cdot \cos \alpha$$

$$\cos(\alpha + \beta) = \cos \alpha \cdot \cos \beta - \sin \alpha \cdot \sin \beta$$

$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = R \begin{bmatrix} x_1 \\ y_1 \end{bmatrix} =$$

$$= \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \end{bmatrix}$$

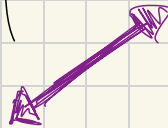
$$\begin{bmatrix} x_2 \\ y_2 \end{bmatrix} = \begin{bmatrix} x_1 \cos \theta - y_1 \sin \theta \\ x_1 \sin \theta + y_1 \cos \theta \end{bmatrix}$$

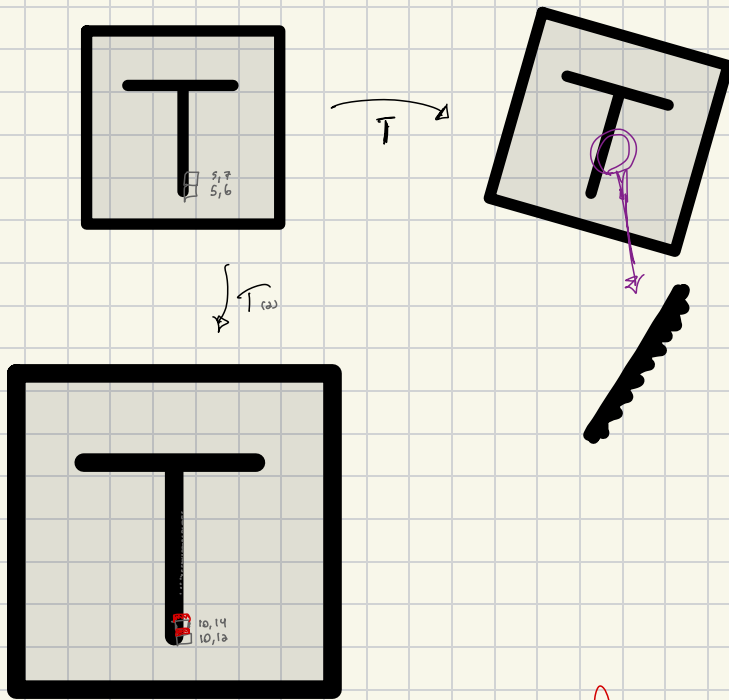
$$x_2 = r \cdot \cos(\phi + \theta) =$$

$$= r (\cos \phi \cdot \cos \theta - \sin \phi \cdot \sin \theta)$$

$$= r \cos \phi \cos \theta - r \sin \phi \sin \theta$$

- $x_2 = x_1 \cdot \cos \theta - y_1 \cdot \sin \theta$
- $y_2 = x_1 \cdot \sin \theta + y_1 \cdot \cos \theta$





★ Interpolation

→ novos pixels
nos
vazios

utiliza dados
da imagem
original

- Objetivos: estimar valores de intensidade em posições desconhecidas

① Interp. por vizinhos mais próximos

$$v(x, y) = f([x'], [y'])$$

□ comp. eficiente

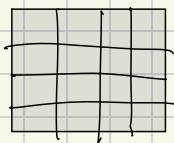
□ produz artefatos → distorção na border

② Int Bilinear

$$v(x, y) = ax + by + cxy + d$$

□ melhora no resultado, + aumento no tempo computacional

③ Int. Bicubica



$$v(x, y) = \sum_{i=0}^3 \sum_{j=0}^3 a_{ij} x^i y^j$$

→ 16 coeficientes

□ Preservar os detalhes finos

□ Custo computacionalmente

